



MONASH University

Assessment

Faculty of Information Technology

FIT5125

IT Research and Innovation Methods Semester 1 2026

Assignment 2

Value

This assignment is worth **45%** of the total marks for FIT5125

Assignment due date

Monday, 8 June 2026, 11:55 PM

Submission method

Submit to Moodle:
three PDF documents, one for each of the three tasks (A, B and C).

Assignment Criteria

Task A	Task B	Task C
Descriptive Statistics	Inferential Statistics	Design
Weighting 12 Marks	Weighting 12 Marks	Weighting 21 Marks
See Page 3	See Page 5	See Page 7

This is an individual assessment; it must be your own work and expressed in your own words.

☐ *A marking guide is available on Moodle.*

☐ *There are specific requirements for file names on your submission (see the instructions for each task).*

Assessment rules:

1. Note that plagiarism detection procedures may be applied to each submission. See the University rules and regulations regarding plagiarism and resulting penalties. Any case of plagiarism detected will mean **automatic failure of the entire assignment**. Note that even where Turnitin reports a high similarity score, this may simply be the result of text that is part of the original question or a correctly referenced quotation (such cases are not plagiarism).
2. Late submissions will incur a penalty of 5% per day, see:
<https://publicpolicydms.monash.edu/Monash/documents/1935752>
3. Submissions should follow the instructions for each task, described under "What to Submit".
4. Monash policy on Special Consideration is available at:
<https://www.monash.edu/exams/changes/special-consideration>
5. We aim to mark and return work within 10 working days of submission.

Task A

"Telling a Data Story"

Telling a coherent story with your data is a core part of the research process. Statistics can be used in myriad ways to describe any given dataset, so it is important to use the appropriate measures and visualisations to enrich and provide context to a narrative of your data. Although in a perfect world we would ask a question and then design a process to capture data that answered that question, we often must make use of 'secondary' data, or data captured previously.

In this assignment, you will tell a story about some public data using the methods you have learned about.

Complete the following:

1. Identify a data source from one of the following open data repositories:

<https://www.data.vic.gov.au/>

<https://www.abs.gov.au/statistics/>

<https://uis.unesco.org>

<https://ec.europa.eu/eurostat/databrowser/>

<https://opendata.cityofnewyork.us/>

2. Formulate a research question that you want to ask of this data – for example "What is the relationship between the value of imported goods and cost of living?" (**max. 50 words**)
3. Select 2 appropriate descriptive metrics (e.g. mean) that tell you something about the data, and calculate those metrics on your selected data, using a tool of your choice. Present the fields from the dataset used and the resulting calculated values (**max. 50 words**)
4. Create an appropriate visualisation to help explain what you are saying about the data. The visualisation should be fully annotated.
5. Write a short narrative description of your findings as they relate to your research question, referencing both your chosen metrics and visualisation. (**max. 200 words**)

What to Submit

A PDF document, named "STUDENT-ID-Assignment-2-Task-A.pdf" containing your response to the assignment.

Use the following headings in your document:

- 0. Your Name & Student ID
- 1. Data Source (list the data you've selected and its source URL)
- 2. Research Question (<50 words)
- 3. Descriptive Metrics Used (< 50 words)
- 4. Annotated Data Visualisation
- 5. Narrative Description (< 200 words)

How Much to Write

There is a strict word limit for your answers detailed above. For submission over the specified word limits, only the parts of answers within the word limit will be awarded marks.

What to Know

If you have any questions about the assignment, you should submit them as a public post to the Ed forum (under the subcategory "Assignments") so that all students have access to the question and CE or admin tutor's response.

This is an individual exercise that forms part of the formal assessment for the unit, you must therefore work alone and follow Monash University's policies, procedures and regulations relating academic integrity, plagiarism and collusion (see Moodle).

As this is a formal assessment tutors are not allowed to support directly, however, they can provide feedback on any studio activities (during the studio).

This task supports unit learning outcome 1 (Demonstrate the ability to frame and design all aspects of the research and innovation pipeline) and unit learning outcome 4 (Evaluate, select and justify research and innovation methods, including data collection and analysis using qualitative and quantitative methods).

Task B

"Working with Hypotheses"

In this exercise you will formulate a hypothesis, prepare a plan of your study (including statistical testing) and justify it, including the potential limitations of it. The topic is **"What attracts young people in Australia to Vaping?"** (Here "young people" means anyone under 25) Imagine you are asked to develop this research area further.

Complete the following:

1. Propose a hypothesis. It should be something you can realistically test using one or more of the statistical tests covered in this unit. It can concern any subtopic or natural phenomena which relates in some way to the topic above. **(max. 50 words)**
2. Write down the null hypothesis. **(max. 50 words)**
3. Write down the independent and dependent variables as well as at least three confounding variables. **(max. 50 words)**
4. Imagine you had a budget of up to \$1,000AUD (in addition to up to 100 hours of your time to conduct the study). Explain what data you will collect to investigate this hypothesis and how you would obtain the data in a practical fashion. **(max. 100 words)**
5. What statistical test(s) do you expect to conduct to test your hypothesis? Provide your assumptions of the data and why such test(s) are appropriate. **(max. 150 words)**
6. What are the limitations of your study? Write a paragraph that explains these limitations as well as potential future investigations you might conduct. **(max. 200 words)**
7. Present a different narrative that could have been presented from your results, highlighting how the data needs to be selected, methods changed, or process otherwise manipulated to support this different interpretation. **(max. 50 words)**

What to Submit

A PDF document, named "STUDENT-ID-Assignment-2-Task-B.pdf" containing your response to the assignment.

Use the following headings in your document:

- 0. Your Name & Student ID
- 1. Proposed Hypothesis (< 50 words)
- 2. Null Hypothesis (<50 words)
- 3. Independent, Dependent and Confounding Variables (< 50 words)
- 4. Data Collection (< 100 words)
- 5. Statistical Tests (< 150 words)
- 6. Limitations (< 200 words)
- 7. Alternative Narrative (< 50 words)

How Much to Write

There is a strict word limit for your answers detailed above. For submission over the specified word limits, only the parts of answers within the word limit will be awarded marks.

What to Know

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Task C

"Design for Emotional Awareness in Online Collaborations"

Design is central to improving how people experience online collaboration. In this task, you will analyse a set of user quotes on Moodle that describe emotional challenges people face when using Zoom. These insights should then be synthesised into clear **design problem statements** that capture where emotional cues are lost, misread, or difficult to express during online interactions.

Using insights from the quotes, you must **identify and describe key emotional issues** experienced by users in online collaborations. Based on this analysis, you will propose **potential design solutions** that address these issues and enhance emotional awareness in Zoom. Your ideas should help users better perceive and communicate emotions, reduce ambiguity and support more inclusive, natural interaction for participants from diverse backgrounds.

For this activity, affinity diagramming, the design problem statements, design rationales, and the annotations to the Miro boards that you submit **must be your own work**.

Part C1: Understanding the Design Context

1. Review the bank of quotes provided on Moodle. These quotes are accounts of user experiences with video collaboration platforms, with particular focus on multilingual contexts.

2. Using **Miro**, develop an *affinity diagram* to identify and synthesise key themes emerging from the provided quotes that relate to emotional issues experienced by users in online collaborations. Begin by coding the excerpts prioritising **latent coding over semantic coding** wherever possible to capture not just what participants said, but the underlying meanings, assumptions, and tensions shaping their experiences. Group similar ideas, patterns, and user sentiments together to uncover the main emotional challenges, needs, frustrations, and opportunities expressed by users in their experiences with online interaction. (This clustering will support the development of **coherent, higher-level themes** grounded in the deeper interpretive insights revealed through latent coding.)

3. Based on the themes you identify, develop two or more **design problem statements** that you will use to brainstorm solutions.

Part C2: Design Ideation

4. Using **Miro**, brainstorm solutions to address your problem statements, consider:

- ◆ What extensions to the features or interface elements on Zoom would help?
- ◆ How would these features or interface elements work?
- ◆ How do they address the needs expressed in the user quotes?

5. Identify two **significantly different** solutions; extend the description of these designs on your Miro board and separately write a design rationale statement for each to capture:

- ◆ How the design addresses the needs and challenges identified in the user quotes.
- ◆ What aspects does your design aim to support or improve?
- ◆ Potential limitations or trade-offs in your design.

What to Submit

A PDF document, named "STUDENT-ID-Assignment-2-Task-C.pdf" containing your response to the assignment.

Use the following headings in your document:

- 0. Your Name & Student ID
- 1. MIRO Board links (Links to annotated copies of your Miro boards for Part 1 and Part 2; annotations should allow a marker to understand your affinity diagramming and brainstorming process. Make sure the links are accessible!)
- 2. Design Problem Statements (<50 words per statement)
- 3. Design Rational Statements (two rationales, < 400 words per rational)

How Much to Write

There is a strict word limit for your answers detailed above. For submission over the specified word limits, only the parts of answers within the word limit will be awarded marks.

What to Know

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This task supports unit learning outcome 6. Communicate the outcomes of research and innovation activities, demonstrating an understanding of diverse audiences and stakeholders.